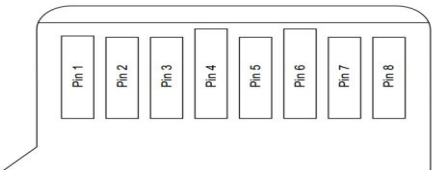
Feature summary

- Highly-integrated memory controller
  Fully compliant with SD Memory Card specification 2.0 and Memory Card specification 2.0 addendum Four integrated 4KByte Sector Buffers for fast data transfer
- Standard Memory Card form factor
  15.0mm x 11.0mm x 1.0mm
- 2.7…3.6V normal operating voltage
- 2.0…3.6V basic communication (CMD0, 15, 55 ACMD41) voltage
- Low-power CMOS technology
- Wear Leveling: equal wear leveling of static and dynamic data.
  The wear leveling assures that dynamic data as well as static data is balanced evenly across the memory. With that the maximum write endurance of the device is guaranteed.
- Patented power-off reliability
  No data loss of older sectors
  Max. 16 sectors data loss (old data kept) if power off during writing
- High reliability
  Best available SLC NAND Flash technology
  Designed for embedded market
  MTBF: 3,000,000 hours
  Durability: 10,000 mating cycles
  Operation Temperature range: 0°C up to 70°C
  Storage Temperature range: -25°C up to 85°C
- Hot swappable
- CPRM keys optional
- High performance
  SD burst up to 15MB/s
  SD Low speed 0…25MHz clock rate
  SD High speed 25…50MHz clock rate
  Up to 20/15 MB/s sustained read/write speed
  Flash burst up to 20MB/s
  “Default speed cards” with disabled high speed mode available
- Available densities
  up to 512MBytes
- Controlled BOM
- Life Time Monitoring SD/SPI with standard or vendor commands

Pin Assignment



Pin No.	SD Mode		
	Name	Type	Description
1	Dat2	I/O/PP	Data Line [Bit 2]
2	CD/DAT3	I/O/PP3	Card Detect / Data Line [Bit 3]
3	CMD	PP	Command/Response
4	VDD	S	Supply voltage
5	CLK	I	Clock
6	VSS	S	Supply voltage ground
7	DAT0	I/O/PP	Data Line [Bit 0]
8	DAT1	I/O/PP	Data Line [Bit 1]

S: power supply; I: input; O: output; PP: I/O using push-pull drivers

Table 1: Pin Assignment

Product List

Part Number	Capacity	Actual Size（Min）（note1）	Speed Class	Write Speed (Typical)	Type
SDHFSAD064M	64MB	58MB	C4	4 MB/S	SD
SDHFSAD128M	128MB	116MB	C4	4 MB/S	SD
SDHFSAD256M	256MB	230MB	C4	4 MB/S	SD
SDHFSAD512M	512MB	460MB	C4	4 MB/S	SD

Table 2：Product List

Note1：*1Gigabyte (GB) = 1024MB=1 billion bytes. Some capacity not available for data storage .

***Maximum speed differs from the bus I/F speed. It varies depending upon the card performance. The average speed that a device writes to an SD memory card may vary depending upon the device and the operation it is performing. Normal and high-speed cards can also be used with host devices, but the high performance enabled by a host device can only be achieved with a memory card.**

Mechanical form factor as follows: (Unit in mm)

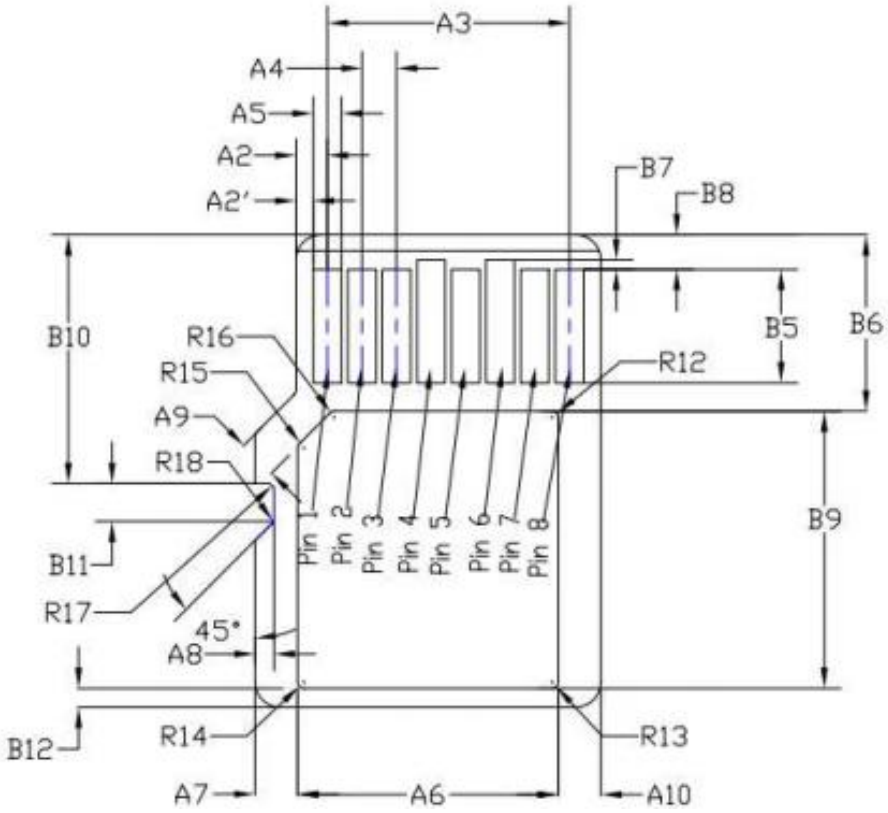


Figure 1: Bottom View

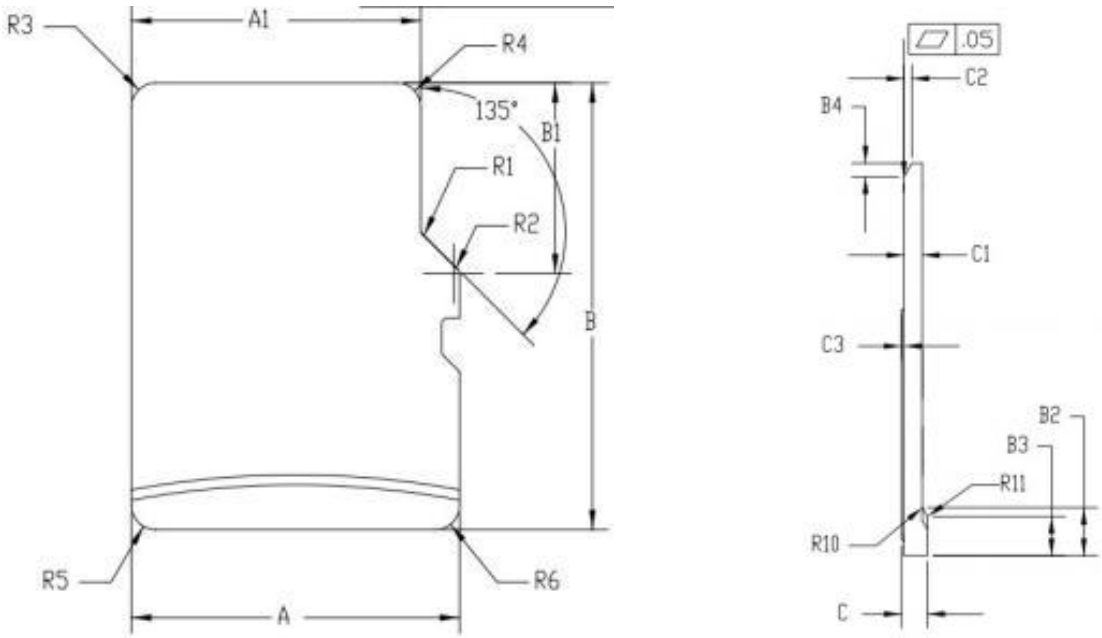


Figure 2: TOP View

Table 3. Physical Dimension Specifications (Unit in mm)

COMMON DIMENSIONS				
SYMBOL	MIN	NOM	MAX	REF.SHEET
A	10.90	11.00	11.10	1
A1	9.60	9.70	9.80	1
A2	0.90	1.00	1.10	2
A2"	0.425	0.550	0.675	2
A3	7.60	7.70	7.80	2
A4	1.05	1.10	1.15	2
A5	0.85	0.90	0.95	2
A6	8.10	8.30	8.50	2
A7	–	–	1.88	2
A8	0.50	0.60	0.70	2
A9	0.80	–	–	2
A10	–	–	1.50	2
B	14.90	15.00	15.10	1
B1	6.30	6.40	6.50	1
B2	1.74	1.84	1.94	1
B3	1.40	4.50	1.60	1
B4	0.42	0.52	0.62	1
B5	3.50	3.60	3.70	2
B6	5.50	–	–	2
B7	0.20	0.30	0.40	2
B8	1.00	1.10	1.10	2
B9	8.60	8.80	9.00	2
B10	7.80	7.90	8.00	2
B11	1.10	1.20	1.30	2
B12	-	-	0.89	2
C	0.90	1.00	1.10	1
C1	0.65	0.70	0.75	1
C2	0.20	0.30	0.40	1
C3	0.00	0.10	0.20	1

COMMON DIMENSIONS				
SYMBOL	MIN	NOM	MAX	REF.SHEET
R1	0.10	0.20	0.30	1
R2	0.10	0.20	0.30	1
R3	0.70	0.80	0.90	1
R4	0.70	0.80	0.90	1
R5	0.70	0.80	0.90	1
R6	0.70	0.80	0.90	1
R7	29.90	30.00	30.10	1
R10	0.10	0.20	0.30	1
R11	0.10	0.20	0.30	1
R12	0.10	0.20	0.40	1
R13	0.10	0.20	0.40	1
R14	0.10	0.20	0.40	1
R15	0.10	0.20	0.40	1
R16	0.10	0.20	0.40	1
R17	0.10	0.20	0.30	1
R18	0.10	0.20	0.30	1

Note:

1,Dimensions are in millimeter.

Pin NO.	PIN NAME(SD MODE)	Critical
1	DAT2	Dimensions to be monitored in Production Before Label Attach: A,A",B,B8,B10 After Label Attach: A10,A7,B12,C3
2	CE/DAT3	
3	CMD	
4	VDD	
5	CLK	
6	VSS2	
7	DATA0	
8	DATA1	